DSA Logical Thinking Challenge (Non-Coding)

Duration: 20–30 Minutes

These questions are based on concepts you've already taught:

- Recursion
- Backtracking
- Tree Traversal
- Divide & Conquer
- Strassen's Algorithm
- Dynamic Programming
- Knapsack
- LCS
- LIS

Question 1 – Recursion

You have 5 gift boxes.

Inside every box is another smaller gift box.

The last box contains a chocolate.

To reach the chocolate, what must you do repeatedly?

1. Open all boxes at once
2. Open one box, then the next, then the next
3. Throw away the boxes
4. Start from the smallest box

Explain your answer.

The correct choice is 2 open one box, then the next, then the next. Because the gift boxes are nested inside each other like Russian dolls, you cannot access any of the inner boxes or the chocolate without first opening the outer layers blocking them.

Question 2 – Backtracking

You are in a maze.

You choose a path and after walking for 5 minutes you find a dead end.

What should you do?

1. Stop walking forever
2. Break the wall
3. Go back and try another path
4. Restart the maze

Which DSA concept does this represent?

This scenario best illustrates Backtracking where a recursive method is used to find the solution incrementally

---

Question 3 – Tree Traversal

A school principal wants to meet everyone.

```
    Principal
  /            \
Teacher A Teacher B
```

In what order would the principal meet everyone if using:

Preorder

```
    Principal
  /           \
Teacher A Teacher B
```

Inorder

```
   Teacher A
  /           \
Principal Teacher B
```

Postorder

```
    Teacher B
  /            \
Principal  Teacher A
```

Write the sequence.

Question 4 – Dynamic Programming

Your friend asks:

25 × 25

You calculate:

625

and write it in your notebook.

After 10 minutes another friend asks the same question.

Would you:

1. Calculate again
2. Use the notebook answer

Which Dynamic Programming concept is this?

Memorization

Question 5 – Knapsack

You have a bag that can carry only 6 kg.

Items:

Laptop = 4kg, Value = 50
Book = 2kg, Value = 20
Camera = 3kg, Value = 40

Which items would you carry to get the maximum value?

Explain your choice.

I would carry the laptop and book which perfectly maximizes the total value of 70 while staying in the 6kg limit

---

Question 6 – LCS

Two students write:

Student A = HELLO

Student B = HLO

What is the longest common subsequence?

Why?

The longest common subsequence is “HLO” both students input the characters H, L, O

In the same sequence appearing in the same exact order

Question 7 – LIS

Marks over 6 years:

50, 60, 55, 70, 80, 75

Find the longest increasing sequence.

50,60,70,80

Question 8 – Divide and Conquer

You have a pile of 1000 papers.

Would it be easier to:

1. Search all papers one by one
2. Divide them into smaller groups and search

Which algorithm strategy does this represent?

It would be easier to divide them into smaller groups and search , it is more easy to search and is more manageable

---

Question 9 – Binary Search Thinking

You are searching for page 250 in a 500-page book.

Would you:

1. Start from page 1
2. Open near the middle first

Which algorithm does this resemble?

Open the middle first since we are looking for 250 which happens to me half of 500 , so opening right in the middle will take you directly to your target in one single action and this resembles Binary Search algorithm

---

Question 10 – Dynamic Programming vs Recursion

Imagine solving a math problem.

You solve it once and write the answer on a sticky note.

Next time the same problem appears, you read the sticky note instead of solving again.

Which is better?

1. Recursion
2. Dynamic Programming

Why?

Dynamic Programming is much better as it ensures that u solve a unique problem only once,

Recursion does not have a thing called a “sticky note” and also u will have to solve the same problem again and again.

Bonus Puzzle

You have three doors.

Door A → Treasure

Door B → Dead End

Door C → Monster

You choose Door C and find a monster.

What should you do next?

We should go to the start and choose Door A to find the treasure

Which DSA concept is being used?

Backtracking
